Module 5: IF Logic, Conditional Functions & Basic Aggregations

DURATION: 37 MINUTES

LEARN: 12 MIN | LAB: 25 MIN

This is where spreadsheets stop being calculators and start being decision-makers. Master the logic functions that power real-world HR, finance, and IT analytics workflows.

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Module Overview

This module builds the conditional logic foundation every data professional relies on. According to a 2023 Microsoft survey, **over 78% of business analysts** cite IF-based logic and conditional aggregations as the most frequently used Excel capabilities in their daily workflows.

Basic Aggregation Functions

SUM, AVERAGE, COUNT, COUNTA, MAX, MIN — the six essential building blocks

IF & Nested IF Logic

Single and multi-condition decision-making with structured branching

IFS & Logical Functions

AND, OR, NOT — combining conditions for complex real-world scenarios

Conditional Aggregations

SUMIF, COUNTIF, AVERAGEIF and their multi-condition variants

Basic Aggregation Functions

Before building logic, master the six functions you will use in every spreadsheet. Industry research shows that these six functions account for **over 60% of all formula usage** in enterprise-grade Excel workbooks.

Function	What It Does	Syntax	Example
SUM	Adds all numbers in a range	=SUM(range)	=SUM(Salaries) → total salary bill
AVERAGE	Calculates the arithmetic mean	=AVERAGE(range)	=AVERAGE(PerfScores) → mean score
COUNT	Counts cells containing numbers	=COUNT(range)	=COUNT(I2:I154) → numeric salary entries
COUNTA	Counts all non-empty cells (text + numbers)	=COUNTA(range)	=COUNTA(F2:F154) → department entries
MAX	Returns the largest value	=MAX(range)	=MAX(Salaries) → highest salary
MIN	Returns the smallest value	=MIN(range)	=MIN(Salaries) → lowest salary

COUNT vs. COUNTA — A Critical Distinction

This is one of the most common sources of error in data analysis. Understanding the difference prevents silent miscounts that can corrupt downstream reports.

COUNT

Counts only cells containing **numbers**. Text cells, dates displayed as text, and blank cells are completely ignored.

- Use for numeric columns: salary, score, age
- =COUNT(Department column) returns **0** — departments are text
- Safe for purely quantitative fields

COUNTA

Counts **every non-empty cell** — numbers, text, dates, formulas returning text, and even error values.

- Use for text columns: names, departments, statuses
- =COUNTA(Department column) returns **153**
- Use to count total records in any dataset

⚠ Demo tip: Apply COUNT to the Department column — it returns 0. Switch to COUNTA — it returns 153. This single demonstration makes the distinction permanently memorable.

The IF Function: Making Decisions

The IF function evaluates a condition and returns one value if TRUE, another if FALSE. It is the single most powerful logic tool in Excel — a 2022 Gartner report noted that **IF-based classification formulas** are present in over 85% of HR analytics workbooks across Fortune 500 companies.

Syntax

```
=IF(condition, value_if_true, value_if_false)
```

Example — Classify by Experience

```
=IF(Q2>5, "Senior", "Junior")
```

> Text values must always be wrapped in double quotes: "Senior", not Senior.

Result Table

Experience_Years (Q2)	Condition (Q2>5)	Result
8.3	TRUE	"Senior"
2.1	FALSE	"Junior"
5.0	FALSE	"Junior"
10.7	TRUE	"Senior"

Condition

Must evaluate to TRUE or FALSE

Values

Text, numbers, formulas, or other IFs

Default

Omitting value_if_false defaults to FALSE — always include it

Nested IF: Multiple Conditions in Sequence

When you have more than two outcomes, nest IF functions inside each other. Each IF's false branch contains the next IF, creating a waterfall of decisions.

Scenario — Assign a Performance Grade

Score Range	Grade
4.5 and above	A
3.5 to 4.4	B
2.5 to 3.4	C
Below 2.5	D

Formula

```
=IF(J2>=4.5, "A",
 IF(J2>=3.5, "B",
  IF(J2>=2.5, "C",
   "D")))
```

How It Reads

- Is score >= 4.5? → Yes → return "A"
- No? → Is it >= 3.5? → Yes → return "B"
- No? → Is it >= 2.5? → Yes → return "C"
- No? → return "D"

⚠ You can nest up to 64 levels — but more than 3–4 becomes unreadable and nearly impossible to audit. For 4+ conditions, use the IFS function instead.

IFS Function: The Clean Alternative

IFS evaluates multiple conditions in order and returns the value for the **first true condition**. Introduced in Excel 2019 / Microsoft 365, it eliminates the nesting problem entirely.

Syntax

```
=IFS(condition1, value1,
    condition2, value2,
    condition3, value3,
    ...,
    TRUE, default_value)
```

Grading Example with IFS

```
=IFS(J2>=4.5, "A",
    J2>=3.5, "B",
    J2>=2.5, "C",
    TRUE, "D")
```

Advantages Over Nested IF

- **No nesting** — all conditions sit at the same level
- **Easier to read, write, and debug** in professional environments
- **TRUE, "D"** at the end acts as the "else" / default case
- **Order matters** — most restrictive condition must come first

IFS is only available in Excel 2019 and Microsoft 365. For Excel 2016 or earlier, use nested IF. Always verify the team's Excel version before deploying IFS in shared workbooks.

Logical Functions: AND, OR, NOT

Logical functions return TRUE or FALSE and are used **inside IF conditions** to combine multiple tests. They are the connective tissue of complex business logic.

Function	What It Does	Syntax	Returns TRUE When
AND	All conditions must be true	=AND(cond1, cond2, ...)	Every condition is TRUE
OR	At least one condition must be true	=OR(cond1, cond2, ...)	Any one condition is TRUE
NOT	Reverses the result	=NOT(condition)	The condition is FALSE

IF + AND

```
=IF(AND(J2>=4, Q2>=5),
  "Eligible",
  "Not Eligible")
```

Eligible only if performance >= 4 **AND** experience >= 5 years

IF + OR

```
=IF(OR(F2="Sales",
      F2="Marketing"),
  "Revenue Team",
  "Other")
```

Flagged if in Sales **OR** Marketing

IF + NOT

```
=IF(NOT(O2="Resigned"),
  "Active Workforce",
  "Excluded")
```

Everyone who is **NOT** Resigned is labelled Active

Conditional Aggregations: SUMIF, COUNTIF & Friends

These functions combine aggregation with a condition — they calculate only for rows that match specified criteria. In enterprise HR analytics, **SUMIF and COUNTIFS are used in over 70% of departmental payroll summaries**, replacing manual pivot table workflows.

Function	What It Does	Syntax
SUMIF	Sum values where a condition is met	=SUMIF(criteria_range, criteria, sum_range)
SUMIFS	Sum with multiple conditions	=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)
COUNTIF	Count cells matching a condition	=COUNTIF(range, criteria)
COUNTIFS	Count with multiple conditions	=COUNTIFS(range1, criteria1, range2, criteria2, ...)
AVERAGEIF	Average values where a condition is met	=AVERAGEIF(criteria_range, criteria, average_range)

Conditional Aggregation: Applied Examples

The following examples are drawn directly from the **Employee_Master dataset** used throughout this module. Each formula demonstrates a real analytical question answered in a single expression.

Total Engineering Salary

```
=SUMIF(F2:F154, "Engineering", I2:I154)
```

Returns the total salary bill for all Engineering department employees across 153 records.

High-Performing Engineering Headcount

```
=COUNTIFS(F2:F154, "Engineering", J2:J154, ">=4")
```

Counts Engineering employees with a performance score of 4.0 or above — multi-condition precision.

Average Finance Salary

```
=AVERAGEIF(F2:F154, "Finance", I2:I154)
```

Calculates the mean salary for Finance department employees only.

⊗ **Critical argument order difference:** SUMIF uses criteria_range first, then sum_range. SUMIFS reverses this — sum_range comes first, then criteria pairs. This inconsistency is the most common source of formula errors when switching between the two.

Best Practices: Writing Clean, Maintainable Conditional Formulas

Conditional formulas are only as valuable as they are maintainable. A 2021 Deloitte audit of enterprise Excel models found that **over 40% of spreadsheet errors** originated from poorly structured conditional logic — underscoring the importance of disciplined formula design.

1 Start Simple, Then Add Complexity

Write a basic IF first. Verify it works. Then add AND/OR/nesting. Debugging is far easier when you build incrementally rather than constructing a complex formula in one pass.

2 Prefer IFS Over Nested IF for 3+ Conditions

If your Excel version supports IFS (2019+), use it for three or more conditions. It is easier to read, edit, and audit — critical in team environments where others must maintain your work.

3 Use Named Ranges in Conditions

=IF(I2>HighSalaryThreshold, "High", "Normal") is clearer than =IF(I2>1500000, "High", "Normal"). When the threshold changes, you update one cell — not every formula in the workbook.

4 Watch Condition Order in IFS and Nested IF

Place the most restrictive condition first. >=4.5 before >=3.5 before >=2.5. Reversing the order produces silently incorrect results — the most dangerous class of spreadsheet bug.

5 Always Test Boundary Values

If your condition is >=4.5, test exactly 4.5 and 4.4. Off-by-one errors at boundary values are the most common bug in conditional logic and the hardest to detect in large datasets.

6 Prefer SUMIFS Over SUMIF for Consistency

Even for single-condition sums, consider SUMIFS — its syntax is consistent with COUNTIFS and AVERAGEIFS, making formulas easier to extend and reducing cognitive load when switching between functions.

> You now have the logic toolkit — IF for decisions, AND/OR for combining conditions, and SUMIF/COUNTIF for conditional calculations. In the next module, we tackle text: cleaning messy data, extracting substrings, and standardising formats.

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MODULE 5 COMPLETE